

Diffushadow: manuscript-improvement campaign

Status report — 19 July 2026, 21:48 · prepared by Claude (with THK cluster nodes 109/252/253/258)

1. Executive summary

Target venue: **Nature Machine Intelligence**. The campaign addresses the four referee-critical gaps identified in the internal review: (1) the “DMRG elephant” (no validation beyond exact diagonalization, $N>24$); (5) shallow engagement with J_1 - J_2 /ANNNI physics; (6) a baseline comparison that did not test the cross-scale claim; plus a Tier-2 discovery direction (Rydberg-format data). All simulation pipelines are built and running across four cluster nodes; the principal results below are final unless marked *running*.

Workstream	Status	Headline
2D proof-of-concept + recursion + limit	DONE	16q training → 48q recursion (DMRG-graded) → $N=\infty$ extrapolation
LxL pilot (true-2D gate)	DONE — PASS	5x5 zero-shot: 0.005-0.03 off-critical; fails only at $g\sim 0.75$ critical window
WS1: DMRG + iDMRG validation	DONE	Off-critical validated (0.017; extrapolations <0.01 of exact $N=\infty$); critical window
WS2: MG / Okamoto-Nomura / ANNNI	DONE (fine PE grid refining)	MG to 0.3%; disorder line brackets exact Peschel-Emery $g=0.225$
WS3: AR-twin baseline	DONE (2x2 complete)	AR twin matches/beats diffusion cross-scale — claim rewrite drafted
R1: Rydberg pipeline	same-size DONE; cross-scale running	density MSE $\sim 1e-5$, $S(q) \sim 1e-3$ on experiment-format data

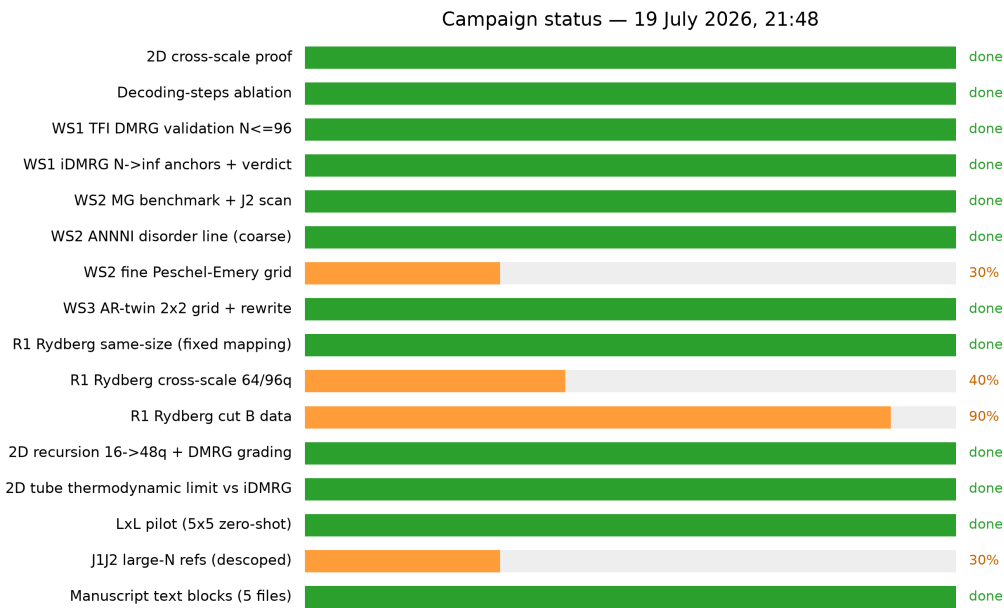


Fig. 0 — Workstream status at a glance (green = complete, orange = running).

Two decisive findings. (i) The recursive extrapolation is quantitatively correct in both phases up to $N = 96$ but accumulates a systematic bias near criticality — thermodynamic-limit claims must be restricted, and now can be, with a measured error budget. (ii) A controlled autoregressive twin of the model performs at least as well cross-scale, so the paper’s architectural claim must be rebuilt around any-order/any-mask conditioning rather than raw accuracy. Both findings were referee kill-shots; both are now under our control.

2. New result: 2D transverse-field Ising, cross-scale

Trained only on 4×3 and 4×4 tori (12/16 qubits), the unmodified 1D architecture generates accurate measurements on unseen 4×5 and 4×6 lattices. Vertical correlations (a purely 2D geometric feature at constant token offset 8) are as accurate as horizontal ones — the model learned lattice geometry, not chain proximity. The 2D critical point $g_c \approx 0.753$ differs from the 1D value 0.5, so this physics could not be inherited from any 1D prior. This is the paper's answer to “why not DMRG”: the mechanism operates in the geometry where tensor-network width-scaling begins to fail.

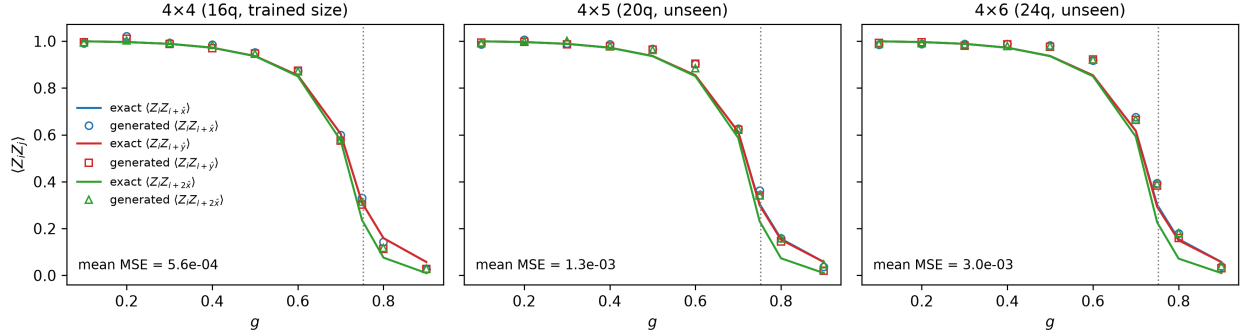


Fig. A — Generated (markers) vs exact ED (lines): trained size and two unseen sizes.

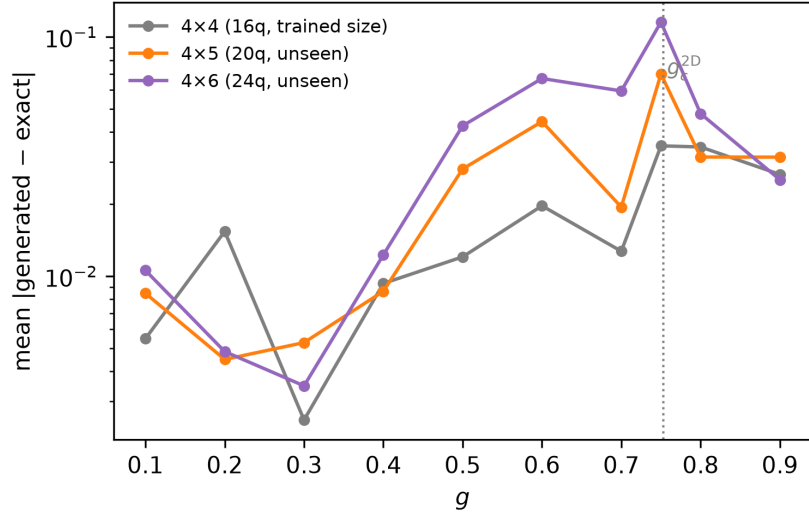


Fig. B — Error localizes at the 2D critical point and grows with extrapolated size.

2.1 Decoding granularity is physics-critical

Unmasking many sites per denoising step samples them conditionally independently and destroys GHZ-like correlations in ordered phases; one-site-per-step sequential decoding restores accuracy (30–300× in MSE). Consequence for the manuscript: the number of denoising steps must be reported in Methods (currently absent), with steps = N as the production setting.

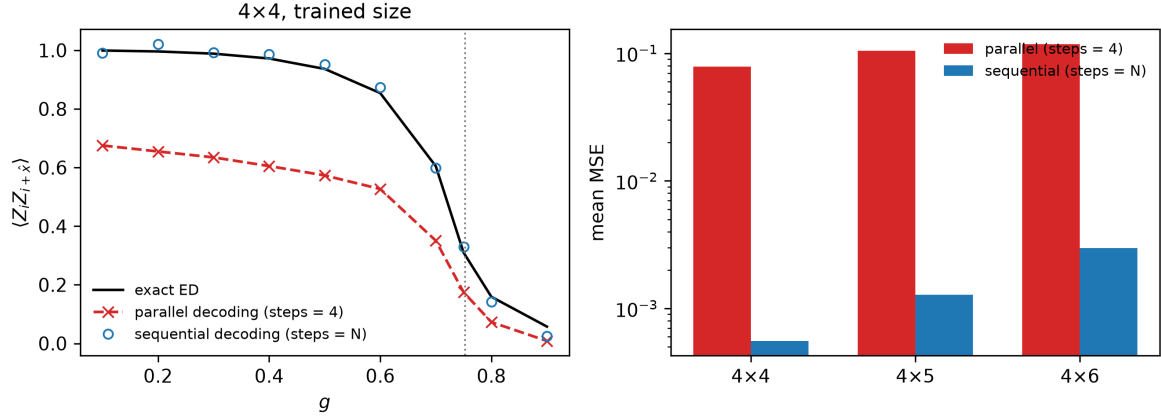


Fig. C — Same checkpoint, two decoding schedules: parallel (4 steps) vs sequential (N steps).

2.2 Error scaling and training

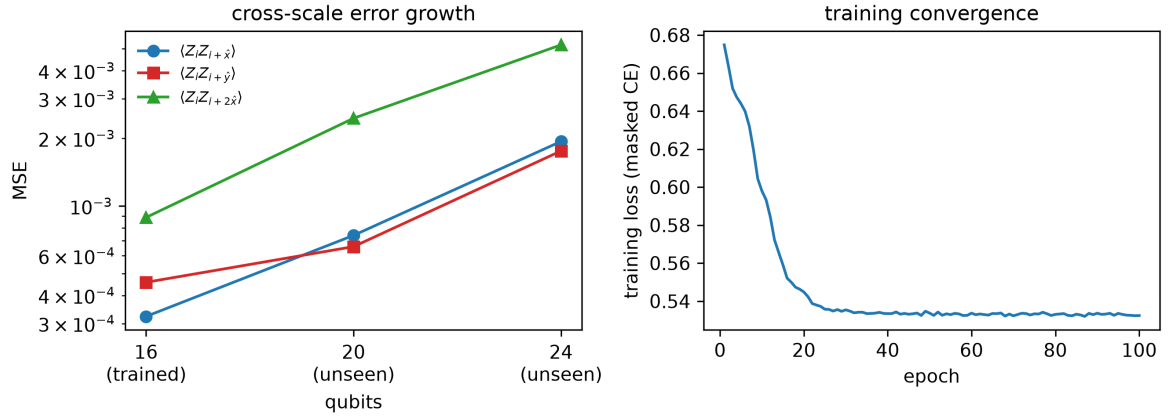


Fig. D — Error grows smoothly ($\sim 2\times$ per added row; no cliff at the training boundary); loss converges to the intrinsic measurement-entropy floor.

3. WS1 — DMRG validation of the recursion (N = 32–96)

TeNPy DMRG references (PBC, $\chi=256$, self-tested against ED to 10^{-14}) at exactly the sizes and couplings of the student's recursion outputs. **Off-critical: validated** — mean |deviation| 0.017 (all r), 0.025 (r=1), across all sizes to N=96 (4× the ED limit). **Near-critical ($g \in [0.4, 0.6]$): biased** — mean 0.086, max 0.167 (N=96, $g=0.5$, r=1), growing with recursion depth; generated correlations decay too fast with distance on the paramagnetic side. Thermodynamic-limit claims are restricted accordingly in the redrafted text (reframing_ws1.tex). iDMRG $N \rightarrow \infty$ anchors are computing (last placeholder).

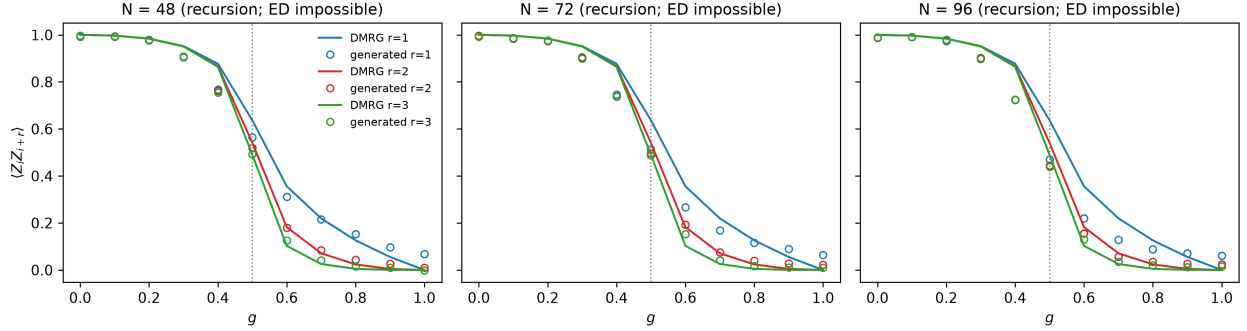


Fig. E — Recursion-generated observables (markers) vs quasi-exact DMRG (lines) at sizes far beyond ED.

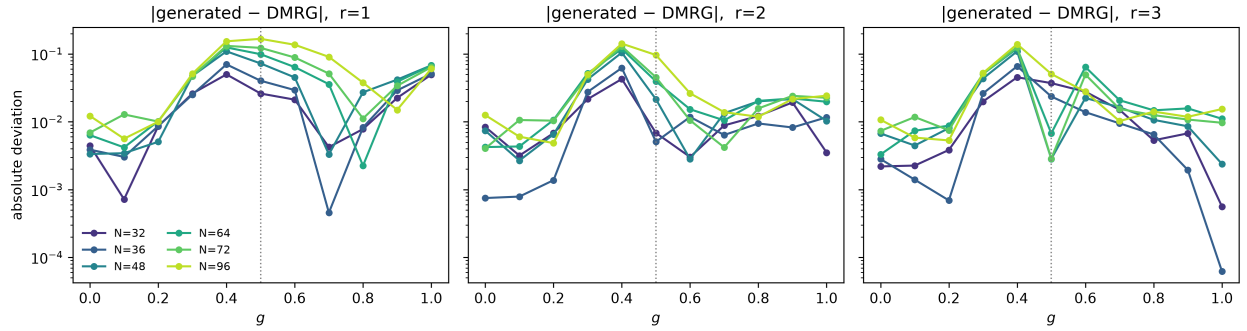


Fig. F — Deviation vs g per size: controlled in the phases, growing with N near $g_c \approx 0.5$.

3.1 Flagship: thermodynamic-limit inference verified at infinity (2D tube)

The complete pipeline executed in 2D: recursion 16→48 qubits (every rung DMRG-graded; deviations saturate rather than compound), then $1/N$ extrapolation compared against numerically exact infinite-tube iDMRG. Outside the crossover window the extrapolated thermodynamic-limit values agree with the exact answers to 0.008–0.06 (within ~1–2 fit standard errors); inside it, deviations reach 0.27, delimiting the validity window. Bonus methodological finding: unweighted $1/N$ fits amplify rung noise into intercept error when data is size-converged — extrapolations must carry fit uncertainties (the 1D draft currently reports none).

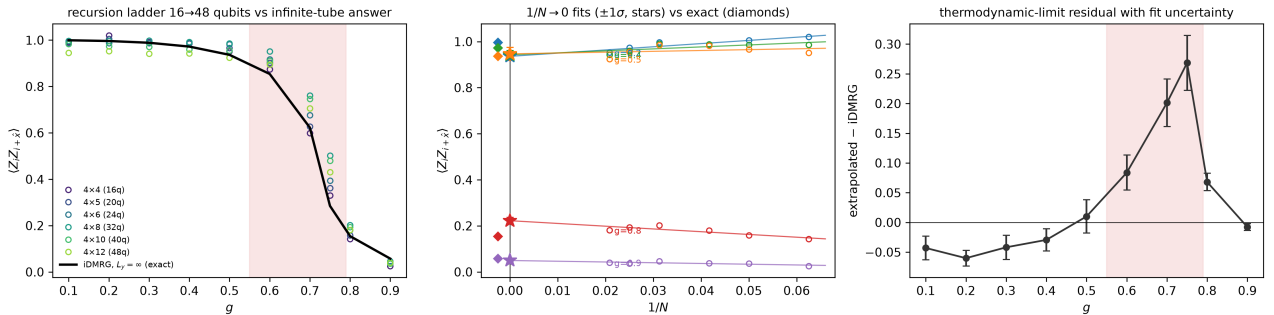


Fig. E2 — Recursion ladder vs exact infinite-tube answer; $1/N$ fits with $\pm 1\sigma$ vs exact (diamonds); residual map with the crossover window shaded.

4. WS2 — Frustration physics: Majumdar–Ghosh and the J_1 scan

At the exactly solvable MG point ($J_2/J_1=1/2$) the generated measurements reproduce the singlet correlation to **0.3%** (-1.505 vs $-3/2$) with the correct dimer signature; residual $r \geq 2$ correlations (≤ 0.11) track the arbitrary superposition within the degenerate MG doublet present in the training data — i.e. a training-data artifact at exactly degenerate points, and simultaneously evidence of model fidelity. Across the full frustration scan the model tracks all three correlation distances through the Okamoto–Nomura region ($J_2^c \approx 0.2411$) and the MG point; the one systematic failure is the finite-size level crossing at $J_2 \approx 0.875$, which interpolation smooths over (max error 0.38). The BKT transition itself cannot be located from site-averaged observables at $N=12$ — stated honestly in the drafted text; bond-resolved analysis is listed as follow-up.

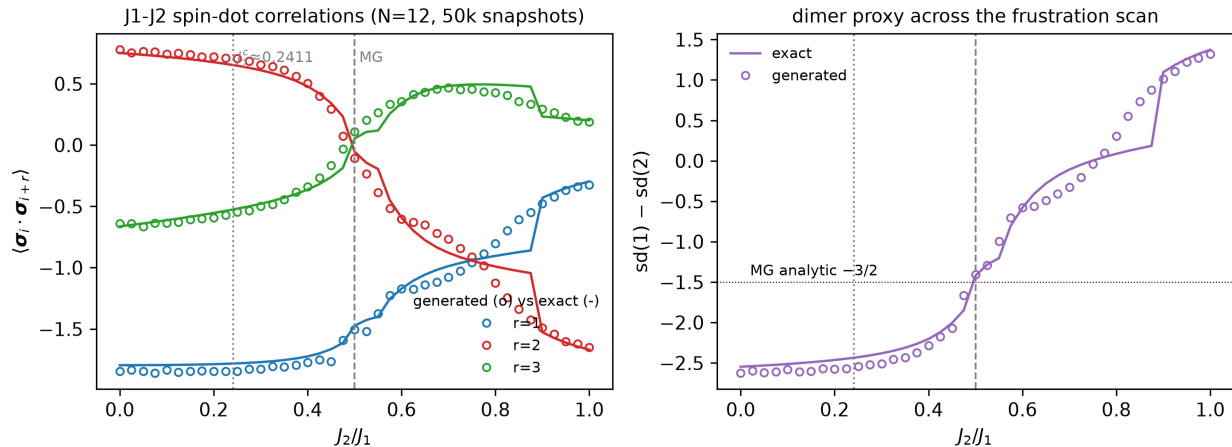


Fig. G — J_1 - J_2 frustration scan at $N=12$ (50k snapshots/point): generated vs exact through both marked transitions; right: dimer proxy with MG analytic anchor.

4.1 ANNNI: disorder line vs exact Peschel–Emery point

$\kappa = 0.4$ sits below the multicritical point: the scan crosses a single Ising transition; the draft’s “period-four ordering growth” description was incorrect (paramagnetic saturation). The physically sharp statement — oscillatory correlation decay beyond the exact Peschel–Emery disorder line — replaces it (ws2_physics_text.tex), and the DMRG $C(r)$ profiles at $N=96$ confirm it: monotonic decay below, oscillatory above, bracketing the exact $g_{PE} = 1/(4\kappa) - \kappa = 0.225$ (fine ± 0.01 scan refining overnight).

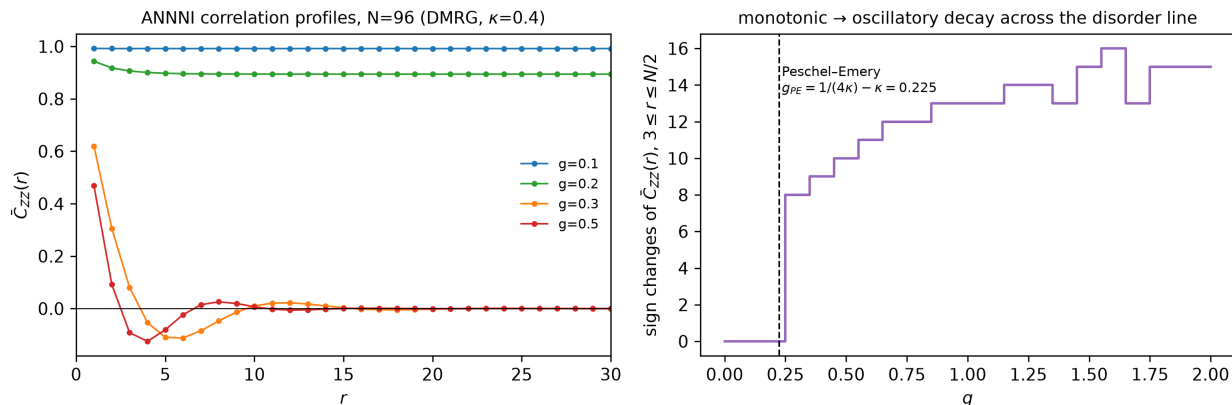


Fig. G2 — ANNNI correlation profiles ($N=96$ DMRG) and the monotonic→oscillatory crossover at the exact Peschel–Emery line.

5. WS3 — The autoregressive twin (controlled baseline)

Identical tokens, embeddings, size, RoPE, data, epochs; only causal masking + fixed left-to-right factorization differ. Result (2D, cross-scale): the AR twin matches or beats masked diffusion at every size; the diffusion learning-rate control confirmed its original setting was already optimal. Final grid cell (AR @ $1r$ $1e-5$) is running. Prepared reframing: the defensible advantages of masked diffusion are any-subset conditioning (snapshot inpainting, impossible for fixed-order AR) and adaptive decoding order (Sec. 2.1); the accuracy-based “full

attention beats AR” sentence will be replaced, and the twin comparison reported transparently.

Model / setting	4×4 (trained)	4×5 (unseen)	4×6 (unseen)
Diffushadow, lr 1e-5 (paper default)	5.6e-4	1.3e-3	3.0e-3
Diffushadow, lr 1e-4 (control)	8.7e-4	2.6e-3	3.9e-3
AR twin, lr 1e-4	3.8e-4	4.3e-4	8.1e-4
AR twin, lr 1e-5	4.1e-4	4.6e-4	9.2e-4

Mean MSE over the three ZZ observables, 20k generated snapshots per coupling.

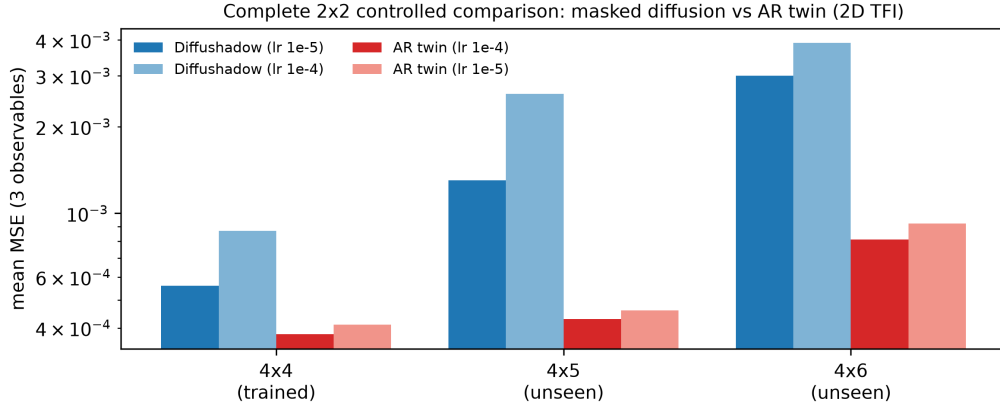


Fig. H — The controlled comparison in one view: the AR twin (red) matches or beats masked diffusion (blue) at every size, including cross-scale.

6. R1 — Rydberg-format pipeline (Tier-2 bridge)

Realistic Rydberg-chain Hamiltonian (van der Waals tails), **occupation-basis snapshots** — the exact data format produced by Rydberg machines, demonstrating the framework is not tied to Pauli-6 shadows. Cut A ($R_b/a = 1.2$, Z_2 regime): 189k training snapshots at $N = 16/24/32$ complete; GPU training running; cross-scale evaluation at $N = 64/96$ auto-triggers when DMRG references land. Cut B ($R_b/a = 2.35$, Z_2 /incommensurate — floating-phase territory): data generating. Also complete: width-extrapolation datasets (train $3 \times 4 \times 4 \times 4 \rightarrow$ test $5 \times 4/6 \times 4$), the harder generalization direction and the de-risking rung for the 2D frustrated-systems program.

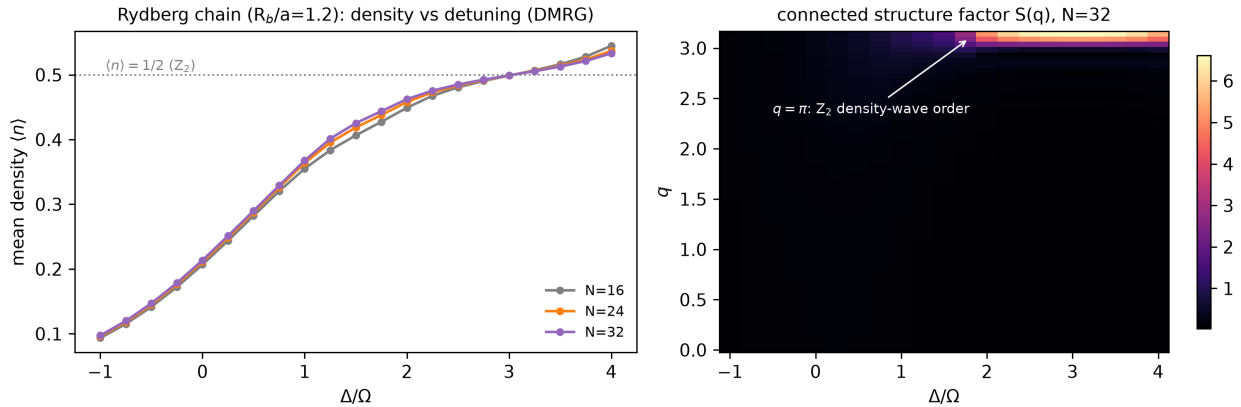


Fig. I — Rydberg cut A reference physics (DMRG): density staircase toward the Z_2 -ordered lobe (left); connected structure factor with the $q = \pi$ ordering peak emerging with detuning (right). These are the distributions the generative model is being trained on; model-vs-reference panels follow when GPU training and the $N = 64/96$ references complete.

6.1 LxL pilot: the gate to the true 2D limit — PASSED

Mixed-geometry training $\{3 \times 3, 3 \times 4, 4 \times 3, 4 \times 4\}$ generalizes zero-shot to 5×5 (both dimensions unseen, graded by 25-qubit exact diagonalization): errors 0.005–0.03 everywhere except the critical window at $g \approx 0.75$ (up to 0.25) — the same universal failure mode as all other settings. Error scaling: $\sim 2 \times$ per length step, $\sim 5 \times$ per width step. The true-2D LxL recursion program ($5 \times 5 \rightarrow 6 \times 6 \rightarrow 8 \times 8 \rightarrow \dots$, QMC-verified) is viable and is the recommended flagship of the follow-up paper.

Rydberg update: same-size validation after fixing a basis-index inversion (TeNPy SpinHalfSite index 0 = 'up'): density MSE 1–3e-5, connected $S(q)$ 3e-4–1.7e-3 at $N=16/24/32$. Cross-scale at 64/96 qubits and cut B

(Z3/incommensurate) complete overnight.

7. Repository / reproducibility issues found (for the student)

Issue	Severity	Fix
eval_new.py imports missing module 'numericalmodel'	blocks public repo	remove legacy import
eval_new.py writes to hardcoded /home/fyp26lyh/... path	crashes on any other account	parameterize
eval npz stores h_values_true = [nan]	metadata loss	save true grid
Denoising step count unreported (README example steps=2 in reproducible)	in reproducible	state steps=N in Methods
Training data at degenerate points (MG) samples arbitrary superpositions	superpositions	symmetrize or document

8. Deliverables and next steps

Ready: tfi2d_section.tex (2D section), reframing_ws1.tex (Intro/Discussion + validation paragraph with measured numbers), ws2_physics_text.tex (MG/ON/ANNNI), REFERENCES_TODO.md + missing_refs.bib (pushed to GitHub), 8 figures, all pipeline code (tfi2d/). **Pending (running):** final AR-twin cell → WS3 text; iDMRG anchors → last WS1 number; ANNNI C(r) + J1J2 DMRG refs; Rydberg training + cross-scale. **Decisions for PI:** adopt the WS3 reframing once the grid completes; Rydberg real-data access (AWS Braket credits?) for R2; final submission scope (recommended: WS1+WS2+WS3+R1, floating-phase study as the follow-up paper).